TECHNOLOGY ETHICS • COMPLETE STUDY NOTES

Set 5: Intellectual Property

Patents, Copyrights, Trademarks, Trade Secrets, DRM, Software Protection Dilemma, DMCA, Open Source & Creative Commons

PART 5

Intellectual property

4 sections, about 2 h 25 min of reading.

IP, protection types, copyright and fair use

~40 min to read | High priority | 8 questions

Source: Set 5, slides 1-9

The Set 5 deck opens with a title slide that lists its four parts: intellectual property, protecting intellectual property, copyright and fair use, and challenges. Its sources are Baase (*A Gift of Fire*) and Quinn (*Ethics for the Information Age*). This section covers the first three parts. The challenges (digital copying, lawsuits, DRM, landmark cases) are in section t13, copyright law and software protection in t14, and open source in t15.

What is intellectual property?

The slide defines intellectual property in one line: **any unique product of the human intellect that has commercial value**. Two ideas carry the definition. The thing must come from a person's mind (it is a creation, not a natural object). It must also be worth money, so that someone would pay to use or copy it.

The slide gives three groups of examples. Learn them, because examples earn marks in a 2-mark definition question.

- Books, songs, movies.
- Paintings, drawings.
- Inventions, chemical formulas, computer programs.

The last line of the slide is a rule: **intellectual property is not physical property**. Physical (tangible) property is a thing you can touch, such as land, a car or a machine. Only one person can hold or use it at a time, and if it is stolen the owner no longer has it. Intellectual property is an idea or a creation. It can be copied without taking anything away from the owner, many people can use it at once, and the same song can sit on a million devices. What the owner owns is the right to control copying and use, not a physical object. Buying a CD gives you the disc but not the song's copyright.

Point	Tangible (physical) property	Intellectual property
Form	Physical thing you can touch: land, buildings, machinery	Creation of the mind: a song, a design, a program, a formula
Use by others	One user at a time; use by one excludes others	Many can use at once; copying leaves the original intact
Theft	The owner loses the item	The owner keeps the original but loses control and income
Protection	Property law, locks, ownership documents	Trade secret, trademark, patent, copyright law, for a limited time

Q: **QB D1 · 2 marks** How is intellectual property different from tangible property?

Q: **CAT 1 · Q2a · 2 marks** What do you understand by the term intellectual property?

Protecting IP: two competing goals

The slide says protection has to serve **two competing goals**.

1. **Benefiting society.** Society wants new inventions and works to reach the **public domain** (free use by everyone) **without delay**. The sooner anyone can use a medicine formula or a program, the more people gain.
2. **Rewarding the creators.** Creators must be paid for their work so that **future inventions are promoted**. If anyone could copy a work the day it appeared, no one would spend years creating it.

The goals pull against each other. Total secrecy and permanent ownership favour the creator and starve society. Instant free copying favours society now but kills the incentive to create. The slide gives the compromise: **the government grants limited ownership rights to IP creators**. "Limited" is the key word. The creator gets a monopoly for a fixed time (for example 20 years for a patent), and afterwards the work passes to the public domain.

The slide then names the protection types. It prints "Trade secrets, Trade secrets, Patents, and Copyrights", which is a typing slip. The four types are **trade secrets, trademarks, patents and copyrights**, and the next four headings follow that order.

> **Key point.** When asked why governments protect IP, name both goals and the compromise: *benefit society*, *reward creators*, so government grants *limited ownership rights*. Naming only the reward side is half an answer.

MARKER WORDS: INTELLECTUAL PROPERTY AND THE TWO GOALS

Say it like this:

> Intellectual property is **any unique product of the human intellect** that has **commercial value**, and it is **not physical property**. Because society wants **benefit** and creators want **reward**, the government grants **limited ownership rights** to creators.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
unique product of the human intellect	defining IP	**SHOULD**
commercial value	defining IP	**SHOULD**
tangible / physical property	contrasting IP with physical property	**MUST**
exclusive rights	saying what protection gives the owner	**MUST**
limited ownership rights	explaining why governments protect IP	**SHOULD**
benefiting society / public domain	stating the first competing goal	**MUST**
rewarding the creators	stating the second competing goal	**SHOULD**

▶ **Recall drill: write the definition from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Define intellectual property.
Any unique product of the human intellect that has commercial value.

The slide's examples of IP (three groups).
Books, songs, movies; paintings, drawings; inventions, chemical formulas, computer programs.

The two competing goals of IP protection.
Benefiting society (inventions reach the public domain without delay); rewarding the creators (promote future inventions).

The solution to the two competing goals.
The government grants limited ownership rights to IP creators.

The four protection types.
Trade secrets, trademarks, patents, copyrights.

Trade secrets

A **trade secret is a confidential piece of IP that provides a company with a competitive advantage**. It works only while it stays secret. Examples on the slide:

- the formula for Coca-Cola syrup;
- the internal design of a system;
- customer lists.

The protection is unusual because the law does not do the work alone. The owner **must take active measures to keep the secret from being discovered**. The slide lists the measures:

- **Locked boxes** (physical security for documents and samples).
- **Non-disclosure agreements (NDAs) and non-compete clauses.** An NDA binds an employee or partner not to reveal the secret. A non-compete clause stops a worker from joining a rival and using what they learned.
- **The Uniform Trade Secrets Act (UTSA).** It imposes **civil liability for misappropriation** of trade secrets, meaning a person who steals or wrongly uses a secret can be sued for damages.

Unlike a patent, a trade secret has no filing, no fixed term and no public description. It lasts as long as it stays secret. If a rival works the formula out independently, or the owner is careless, protection is gone.

Trademarks

A **trademark is a word, a logo, a sound, etc., that identifies a company or a product**. Its job is to tell buyers who made the goods, so a customer can tell the real product from an imitation.

Protection: the company can **sue for improper use** of its trademark. There is a risk, though. If the trademark name **becomes a common noun**, the trademark may be lost, because the name no longer points to one company. The slide's examples are **aspirin, escalator, thermos, xerox and yo-yo**. These were once brand names and became ordinary words for the whole class of product.

> **Watch out.** A trademark protects a name or sign, not an invention or a creative work. Do not write that a trademark protects a program's code. That is copyright.

Patents

A **patent is a public document that provides a detailed description of an invention**. In return for publishing the details, the owner gets an **exclusive right to the invention**. The owner can **prevent others from making, using or selling the invention for 20 years**. After 20 years the invention is in the public domain and anyone may use it.

This is the clearest case of the compromise in the two-goals slide. Society receives the full description of the invention immediately (it is public), and the inventor receives a monopoly for 20 years. Compare it with a trade secret: a patent is public and lasts 20 years, while a trade secret is hidden and can last forever.

beyond the slides To be patented an invention must normally be novel, non-obvious and useful, and the 20 years run from the date the application is filed. The past-paper solutions give these conditions.

Type	Protects	How protected	Slide example
Trade secret	Confidential IP that gives a competitive advantage	Owner's active measures (locked boxes, NDA, non-compete); UTSA civil liability	Coca-Cola syrup formula, customer lists
Trademark	Word, logo or sound that identifies a company or product	Company can sue for improper use; lost if the name becomes a common noun	aspirin, escalator, thermos, xerox, yo-yo
Patent	An invention, described in a public document	Exclusive right: others cannot make, use or sell it for 20 years	An invention
Copyright	An original work	Five rights for the owner; length set by copyright law	Book, song, program

Q: **CAT 1 · Q2b · 6 marks; also 2017 Q2D · 6 marks** **Differentiate between the different available methods of intellectual property protection.**

Q: **2014/15 Q5c; also 2011/12 and 2013/14 papers · 3 marks** **Identify three (3) ways in which an engineer can protect intellectual property.**

Q: **2016 Q4B · 6 marks** **What two forms of legal property protection are most frequently applied to software? List two or three of their key characteristics.**

MARKER WORDS: THE FOUR PROTECTION TYPES

Say it like this:

> The four types are **trade secrets, trademarks, patents and copyrights**. A trade secret is a **confidential piece of IP that provides a company with a competitive advantage**, a trademark is a **word, logo or sound that identifies a company or product**, and a patent is a **public document** that gives the owner an **exclusive right** to the invention for **20 years**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
competitive advantage	defining a trade secret	**MUST**
non-disclosure agreements, non-compete clauses	saying how a trade secret is kept	**SHOULD**
misappropriation / civil liability (UTSA)	saying what law backs trade secrets	**SHOULD**
identifies a company or product	defining a trademark	**MUST**
becomes a common noun	explaining how a trademark is lost	**SHOULD**
detailed description of the invention	defining a patent	**SHOULD**
exclusive right ... 20 years	stating what a patent gives	**SHOULD**
making, using, or selling	saying what a patent owner can prevent	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Define a trade secret.
A confidential piece of IP that provides a company with a competitive advantage.

Three examples of trade secrets.
The formula for Coca-Cola syrup; the internal design of a system; customer lists.

Protections for a trade secret.
Owners must take active measures to keep it from being discovered: locked boxes; non-disclosure agreements (NDA) and non-compete clauses; the Uniform Trade Secrets Act (UTSA), which imposes civil liability for misappropriation.

Define a trademark.
A word, a logo, a sound, etc., that identifies a company or a product.

Trademark names that became common nouns.
Aspirin, escalator, thermos, xerox, yo-yo.

Define a patent, and what it gives the owner.
A public document that provides a detailed description of an invention. It gives the owner an exclusive right to the invention: the owner can prevent others from making, using, or selling it for 20 years.

Copyright and its five rights

Copyright **provides the owner of an original work with five rights**. Learn the list whole and in order, because it is a standard marker list.

1. **Reproduction** (making copies).
2. **Distribution** (selling, lending or giving out copies).
3. **Public display** (showing the work in public, for example a painting or an image on a screen).
4. **Public performance** (playing or performing the work in public, for example a song or a film).
5. **Production of derivative works** (new works built on the original, such as a translation, a film of a novel or a modified version of a program).

The **length of the protection is defined by copyright laws**, and the slide adds that these laws **have been expanded greatly over the years**. The exact terms and the history of the extensions are in t14.
beyond the slides Copyright protects the expression of an idea (the actual song, the actual code), not the idea itself, and it arises automatically when the work is created.

Anyone who does one of these five things without the owner's permission infringes the copyright, unless the use is a fair use.

Q: CAT 1 2021-22 solution file · Q3c · marks not stated **Describe the key differences between intellectual property rights and copyrights.**

Q: Solution file · CAT 1 2021-22 Q4a · marks not stated **What justifies the need of a copyright policy in the Engineering profession?**

Fair use

Copyright does not give the owner total control. The slide says: **under some circumstances it is legal to reproduce a copyrighted work without permission**. This is **fair use**. It is how the law protects society's side of the two-goals trade-off, for example for teaching, criticism and research.

There is no fixed list of allowed copies. **Courts consider four factors**, and the slide gives a guide for each:

Factor	What the court asks	Slide guide
1. Purpose and character of use	Why was it copied? Teaching, criticism, private study, or profit?	Educational use is generally OK
2. Nature of the work	What kind of work was copied?	Nonfiction and non-arts work are more permissible
3. Amount of the work being copied	How much was taken?	Small portions are more permissible
4. Effect on the commercial market for the work	Does the copy reduce sales or licence income of the original?	Should have a negligible effect

Reading the guides the other way gives the factors against fair use: commercial use, creative or artistic works (a novel, a game, a film), copying a large portion or the heart of the work, and a copy that replaces a sale.

The slide's two examples

- **Journal articles on a class website (OK).** A professor scans a few journal articles and posts them on his class website. Students in the class get in with a password. This is fair use: the purpose is educational, journal articles are non-fiction, only a few were copied, and the password keeps the audience to the class, so the market effect is negligible.
- **Painting slides (maybe not OK).** An art professor photographs some paintings from a book as slides and uses them in her lectures. The use is educational, but the works are art (a creative work, so less permissible) and the copy might replace the sale or licence of the images. A court could go either way, which is why the slide says "maybe not OK".

> **Watch out.** Fair use is a defence a court decides case by case on all four factors together. No single factor settles it, and "educational" alone does not guarantee a win.

MARKER WORDS: COPYRIGHT AND FAIR USE

Say it like this:

> Copyright gives the owner of an **original work** **five rights**: **reproduction, distribution, public display, public performance and production of derivative works**. **Fair use** makes it legal to reproduce a copyrighted work **without permission** in some circumstances, and courts weigh **four factors**: **purpose and character of use, nature of the work, amount copied, and effect on the commercial market**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
original work	saying what copyright protects	**SHOULD**
reproduction / distribution	listing the five rights	**MUST**
public display / public performance	listing the five rights	**SHOULD**
derivative works	listing the five rights, or judging a modified copy	**MUST**
without permission	defining fair use	**SHOULD**
purpose and character of use / educational use	applying factor 1	**SHOULD**
nature of the work	applying factor 2 (non-fiction is more permissible)	**SHOULD**
amount of the work / small portions	applying factor 3	**SHOULD**
effect on the commercial market / negligible effect	applying factor 4	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The five rights a copyright gives the owner.
Reproduction; distribution; public display; public performance; production of derivative works.

What is fair use?
Under some circumstances, it is legal to reproduce a copyrighted work without permission.

The four factors courts consider in fair use.
Purpose and character of use; nature of the work; amount of work being copied; effect on the commercial market for the work.

Slide guide for each fair-use factor.
Educational use is generally OK; nonfiction and non-arts work are more permissible; small portions are more permissible; should have a negligible effect on the market.

Worked case: Mofonic, Bunten and Schaeffer (2016 Q6, 20 marks)

This question tests copyright, fair use and (beyond your slides) personality rights. Everything the slides cover is marked as such; the extra law is labelled. The answer is a memo to the judge in three parts, as the marks suggest: A claims (10), B defences (5), C remedies (5).

The facts, retold

- **The game.** Mofonic Games made *Stormcrasher*, a fantasy epic about Thessia, a dragon trainer and mercenary. It runs only on the Degenatron console, has sold over 5 million copies and retails at $50. Players can customise every aspect of Thessia's looks, and the story cutscenes (short high-resolution videos) show Thessia as the player designed her.
- **Bunten.** Roberta Bunten is the voice actress for Thessia. She worked six weeks, eight hours a day, in a studio Mofonic rented. Mofonic's producers directed her and chose the best take. She got a flat fee of $50,000 plus a $100,000 bonus when sales passed one million. Her contract says nothing about who owns the intellectual property.
- **The ending.** Many players disliked the ending, in which Thessia dies. Fan Steve Schaeffer made a new ending. He designed his Thessia to look exactly like Bunten (an uncanny resemblance), then spliced game clips from all over the game, never more than two or three words from any one scene, to make a version where Thessia becomes Supreme Emperor and gets a birthday cake. He posted the video free on fan sites. Fans were split.
- **The sales.** Schaeffer then offered "Stormcrasher: The Supreme Emperor Edition". A player sent him an original game disc and $10, and he mailed back a Degenatron disc carrying the game with his ending. He kept the originals and made coasters and wind chimes of them. He bought 500 blank discs for $1,000 and filled 300 orders, paying $4 postage each. Revenue was 300 x $10 = $3,000. Costs were $1,000 plus 300 x $4 = $1,200, so his profit was about $800.
- Mofonic and Bunten have sued Schaeffer separately. You are law clerk writing the memo.

Q: **2016 Q6 · 20 marks (A 10, B 5, C 5)** **Write a memo to the judge explaining A. the intellectual property claims Mofonic and Bunten can bring (10 marks), B. any defenses Schaeffer can raise (5 marks), C. the remedies, if any, that Mofonic and Bunten might be entitled to (5 marks). [Facts as retold above; reconstructed from garbled scan.]**

MARKER WORDS: APPLYING COPYRIGHT TO A CASE

Say it like this:

> The copier infringes the owner's **rights of reproduction, distribution and derivative works** unless **fair use** applies, which a court decides on the **purpose and character of use, the nature of the work, the amount copied and the effect on the commercial market**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
infringement	naming the wrong done by an unauthorised copy	**MUST**
derivative work	judging a modified version of a game or program	**MUST**
without permission	describing the copy	**SHOULD**
commercial purpose	weighing factor 1 against the copier	**SHOULD**
small portions	weighing factor 3 for the copier	**SHOULD**
negligible effect on the market	weighing factor 4	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Protection challenges, DRM and the landmark cases

~35 min to read · High priority · 8 questions

Source: Set 5, slides 10-19. The definition of IP, the four protection types, copyright and fair use are in section t12. Copyright term history, the DMCA, software copyright and software patents are in section t14.

IP protection challenges: digital copies and peer-to-peer sharing

Copyright, patents and trade secrets were designed for a world of physical copies. Digital technology breaks the assumptions behind them. The slides list the challenges under the heading "IP Protection Challenges". The first is **digital copies**.

- **Easy to make perfect copies of CDs and DVDs.** A photocopy or a cassette copy is worse than the original. A digital copy is identical to it, and a copy of a copy is identical too. Copying costs almost nothing.
- **Easy to download books, music and videos.** Once content is digital, it travels over the network in seconds.
- **Peer-to-peer (p2p) networks allow strangers to share music and other data files.** There is no shop, no publisher and no central owner in between. Any person with a connection can become a distributor.

The result is that the copyright owner's rights of reproduction and distribution (two of the five rights in t12) are exercised by millions of ordinary people, mostly without permission.

The RIAA lawsuits

The music industry body, the **RIAA** (Recording Industry Association of America), reacted by suing. The slides give these facts.

- Since 2003 the RIAA has filed **more than 26,000 lawsuits against individuals** for illegal downloading.
- **Tanya Andersen:** sued by the RIAA, and the case was dropped.
- **Jammie Thomas:** fined **$222K for sharing 24 songs**.

The two names show both sides of the problem. One defendant was sued and the case was dropped. The other received a fine that looks far out of proportion to the value of 24 songs. Many people argue that suing individual users is an unfair and ineffective way to protect IP. It also does nothing about the people who run the sharing networks, which is why the network cases below matter.

beyond the slides Thomas was found liable for infringement at trial. The damages award was later changed on appeal and retrial. For the CAT, quote the slide figure of $222K for 24 songs and move on.

Fair use issues and the software protection question

The second group of challenges on slide 11 is posed as questions, not answers. Treat them as open debates.

- **Fair use issues.** How do we **draw the line** between fair use and infringement? How do we **protect the legal fair use right** of ordinary users when owners add technical locks to content? (The four fair use factors are in t12.)
- **Software protection.** Should software be **copyrighted**? Should software be **patented**? These two questions are answered in t14.

The three answers to the challenges: prevention, tracking, punishing

Slide 12 groups the responses into three words. Memorise them as a set of three, each with its example.

Answer	Meaning	Slide example
Prevention	Make copying technically hard or impossible	Digital IP protection techniques, for example DRMs
Tracking	Find out who copied, and where a copy came from	Digital watermarking
Punishing	Make infringement costly through the law	Extending copyright laws

> **Key point.** The slide adds that this is **an on-going process** and that **some of the changes are controversial**. Use both phrases when you discuss the three answers.

Prevention is covered under DRM further down. Punishing is covered in t14 (longer terms, felonies, minimum damages, the DMCA). Tracking by digital watermark means hiding an identifying mark in the content so that a leaked copy can be traced.

> **Q:** `ethics-cat-1 Q2c · 6 marks` **With the aid of examples, briefly explain why intellectual property protection is much harder to enforce in this internet age.**

> **Q:** `2017 Q2C · 5 marks` **Briefly analyze why intellectual property protection is that much harder to enforce in the internet age. Illustrate your positions with clear examples.**

> **Q:** `2014/15, 2013/14 papers · 6 marks` **How does Internet and other digital technology access undermine intellectual property protection? [reconstructed from garbled scan]**

MARKER WORDS: IP PROTECTION CHALLENGES

Say it like this:

> Digital technology makes it **easy to make perfect copies** and to **download** content, and **peer-to-peer networks** let **strangers share** files, so owners respond with **prevention**, **tracking** and **punishing**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
peer-to-peer network	explaining how strangers share files with no middleman	**SHOULD**
digital rights management (DRM)	naming the technical answer to copying	**MUST**
perfect copies	explaining why digital is different from physical copying	**SHOULD**
illegal downloading	describing what the RIAA sued for	**SHOULD**
prevention, tracking, punishing	listing the three answers to the challenges	**SHOULD**
digital watermarking	giving the example of tracking	**SHOULD**
draw the line	discussing fair use difficulties	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The three answers to the IP protection challenges.
Prevention (digital IP protection techniques, e.g. DRMs); tracking (digital watermarking); punishing (extending copyright laws).

Three digital copy challenges listed on the slides.
Easy to make perfect copies of CDs and DVDs; easy to download books, music and videos; peer-to-peer networks allow strangers to share music and other data files.

RIAA lawsuit figure and the two named defendants.
More than 26,000 lawsuits against individuals since 2003; Tanya Andersen (case dropped); Jammie Thomas (fined $222K for sharing 24 songs).

The two software protection questions.
Should software be copyrighted? Should software be patented?

Significant fair use cases: Sony and Diamond

Slide 13 gives two cases where the court decided that a form of private copying was fair use. They give the practical meaning of the fair use rule in t12.

Sony v. Universal City Studios [1976]

- **Issue:** using a VCR to tape TV programmes for later viewing.
- **Verdict: time shifting is fair use.** The US Supreme Court decided this by 5-4.

Time shifting means recording a broadcast you have a right to watch, so that you can watch it at a more convenient time. The viewer is not creating a new market for the programme. The date on the slide is 1976. beyond the slides That is when the suit began. The Supreme Court ruled in 1984, and the VCR in question was the Sony Betamax. Quote 1976 in the CAT as it is on your slides.

RIAA v. Diamond Multimedia Systems [1998]

- **Issue:** transforming music CDs to MP3 format and playing them on the **Rio** player.
- **Verdict: space shifting is fair use.**

Space shifting means moving content you already own from one medium or device to another, here from a CD to a portable MP3 player. The user has already paid for the music and uses it privately.

> **Watch out.** Do not mix the two up. Sony is the VCR and *time* shifting. Diamond is the Rio MP3 player and *space* shifting. Both were decided in favour of the user.

Reverse engineering cases

Reverse engineering means taking a finished product apart to find out how it works. For software this means **decompiling** it, turning machine code back into readable form. The legal question is whether the copying made during decompiling breaks copyright. Slide 14 lists three cases and gives details for the first.

Sega v. Accolade [1992]

- **Issue:** Accolade made video games to run on Sega game machines. To do so, it **decompiled Sega's game engine**.
- **Verdict:** reverse engineering **does not violate copyright if the intention is to make new creative works, not copy the original work**.

Accolade wanted its own games to be compatible. It did not want to sell Sega's code, so the copying was a means to an end. This is the line to remember: new creative work is allowed, copying the original is not.

Atari v. Nintendo [1992] and Sony v. Connectix [2000]

The slides name these two cases as other significant reverse engineering cases but give no issue or verdict, so list them by name and date with Sega v. Accolade. **beyond the slides** In Atari v. Nintendo, Atari wanted to make games for the Nintendo console but lost, partly because it obtained Nintendo's program code in a way the court found improper. In Sony v. Connectix, Connectix reverse engineered the Sony PlayStation to build software that ran PlayStation games on a PC. The court held that the copying done in the process was fair use, which supports Accolade's result.

Q: **Practice · 4 marks** **Distinguish between time shifting and space shifting, naming the case in which each was held to be fair use.**

Q: **Practice · 4 marks** **What is reverse engineering of software, and when is it lawful? Name one case.**

MARKER WORDS: FAIR USE AND REVERSE ENGINEERING CASES

Say it like this:

> In **Sony v. Universal** the Supreme Court held that **time shifting** is fair use, in **RIAA v. Diamond** that **space shifting** is fair use, and in **Sega v. Accolade** that **reverse engineering** is lawful if the aim is **new creative works**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
time shifting	describing taping TV for later viewing	**SHOULD**
space shifting	describing CD to MP3 on the Rio	**SHOULD**
5-4 Supreme Court	giving the Sony verdict	**SHOULD**
decompiled the game engine	explaining what Accolade did	**SHOULD**
new creative works, not copy the original	giving the Sega v. Accolade rule	**SHOULD**
fair use	naming the legal doctrine behind both shifting cases	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Sony v. Universal City Studios: issue and verdict.
Issue: using a VCR to tape TV programs for later viewing. Verdict: time shifting is fair use (US Supreme Court 5-4).

RIAA v. Diamond Multimedia Systems: issue and verdict.
Issue: transforming music CDs to MP3 format and playing on the Rio player. Verdict: space shifting is fair use.

Sega v. Accolade: issue and verdict.
Issue: Accolade decompiled Sega's game engine to make games for Sega machines. Verdict: reverse engineering does not violate copyright if the intention is to make new creative works, not copy the original work.

The three reverse engineering cases on the slides.
Sega v. Accolade [1992]; Atari v. Nintendo [1992]; Sony v. Connectix [2000].

File sharing cases: Napster

RIAA v. Napster [2001]

- **Issue:** a peer-to-peer music exchange network.
- **Verdict:** web sites **could be liable for users' actions** if they took measures that **deliberately encouraged users to do illegal things**. This is called **secondary infringement**.

Background from the slide. Napster was started by a Northeastern student in the summer of 1999. In two years it had **26 million registered users**. In court, Napster argued that **it did not copy any files**. It said it was **merely providing a directory service**, so the users, not Napster, were the ones copying.

The court did not accept that Napster could shelter behind its users. **Secondary infringement** means being held responsible for someone else's copying. The direct infringers are the users, and the secondary infringer is the service that helps and encourages them. `beyond the slides` Napster kept a central index of which user had which file, which made it easy to shut down and easy to prove it helped.

File sharing cases: Grokster

MGM v. Grokster [2005]

- **Issue:** a **decentralized p2p file-sharing network**.
- **Verdict:** in favour of MGM, by the Supreme Court.

The holding, which you should learn word for word:

> "We hold that one who distributes a device with the object of promoting its use to infringe copyright, as shown by clear expression or other affirmative steps taken to foster infringement, is liable for the resulting acts of infringement by third parties."

Evidence in the case: Grokster advertised "an infringe use" of its service. That advertising was the "clear expression" of intent to promote infringement.

How Grokster differs from Napster: Grokster had no central directory. Its network was decentralized, so the Napster argument "we only run a directory" could not be used. The Supreme Court focused on **intent**, shown by what the company said and did. The words to underline in the quote are **object of promoting**, **clear expression or other affirmative steps**, and **liable for third parties**.

> **Watch out.** The law does not punish a device just because it can be misused. It punishes distributing it *with the object of promoting infringement*. The Napster slide says the same idea: sites are liable if they *deliberately encouraged* illegal acts.

	RIAA v. Napster [2001]	**MGM v. Grokster [2005]**
Network	Peer-to-peer music exchange, directory service	Decentralized p2p file sharing
Defence	"We did not copy any files, only provided a directory"	Not stated on the slides
Verdict	Liable for users' actions if it deliberately encouraged them (secondary infringement)	For MGM; liable for third parties' infringement if it promoted that use
Key evidence	26 million registered users in two years	Advertised "an infringe use"

Q: **Practice · 6 marks** **Explain the Napster and Grokster cases and what they decided about the liability of file sharing services.**

MARKER WORDS: NAPSTER AND GROKSTER

Say it like this:

> Under **secondary infringement**, a service is **liable for the infringement of third parties** if it distributes a device **with the object of promoting its use to infringe copyright**, as shown by **clear expression or other affirmative steps**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
peer-to-peer	describing what Napster and Grokster ran	**MUST**
secondary infringement	naming the liability the Napster court applied	**SHOULD**
directory service	giving Napster's defence	**SHOULD**
26 million registered users	showing the scale of Napster	**SHOULD**
decentralized	describing Grokster's network	**SHOULD**
affirmative steps to foster infringement	quoting the Grokster holding	**SHOULD**
liable for third parties	stating the result of Grokster	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

RIAA v. Napster [2001]: issue and verdict.
Issue: peer-to-peer music exchange network. Verdict: web sites could be liable for users' actions if they took measures that deliberately encouraged users to do illegal things (secondary infringement).

Napster's background and its defence in court.
Started by a Northeastern student in the summer of 1999; 26 million registered users in two years. In court, Napster argued that it did not copy any files; it merely provided a directory service.

MGM v. Grokster [2005]: issue and verdict.
Issue: decentralized p2p file-sharing network. Verdict: in favour of MGM (by the Supreme Court).

The Grokster holding.
"We hold that one who distributes a device with the object of promoting its use to infringe copyright, as shown by clear expression or other affirmative steps taken to foster infringement, is liable for the resulting acts of infringement by third parties."

The evidence against Grokster.
Grokster advertised "an infringe use" of its service.

DRM examples for CDs and DVDs

DRM stands for **digital rights management**. It is the main *prevention* technique from slide 12. It means technical methods built into media, devices or software to control how content is copied and used. Slides 17 and 18 give examples.

DRM for CDs (slide 17)

- **Exploiting a hardware difference.** A CD-ROM drive in a computer and an ordinary CD player read discs differently. The slide calls these the **"Yellow Book"** (CD-ROM) and **"Red Book"** (audio CD) standards. A disc is made to play on a CD player but not to be read properly by a computer, so it cannot be copied easily.
- **Annoying sounds.** Patterns are encoded into the audio so that, when the disc is decoded or copied, they turn into **annoying sounds**. The original plays cleanly and the copy does not.
- **Secure Digital Music Initiative (SDMI).** This introduces a **"digital watermarking"** scheme. A watermark is hidden information in the music that identifies it as protected or marks its source. This is the *tracking* answer from slide 12.

DRM for DVDs (slide 18)

- **Content Scramble System (CSS).** An **encryption scheme** for DVD contents. You need **decryption keys** to view a DVD. **Status: in use, but a crack code exists.**

The CSS case: Jon Johansen and 2600.com

The slide calls this "an important case".

1. **Jon Johansen wrote a decryption program for Linux.** At the time there was no licensed DVD player for Linux, so he wrote one that could decrypt CSS.
2. **2600.com published the code.** The web site put the decryption program online.
3. **Motion picture studios sued 2600.com and won.** The US courts ruled for the studios.
4. **Johansen was tried in Norway and found not guilty.** Norway is where he lived and wrote the code.

The point of the case is that the same act, breaking CSS, was punished in one country (the site that published it) and not in another (the writer). **beyond the slides** The program is generally known as DeCSS. The US case against 2600.com relied on the DMCA anti-circumvention rule covered in t14.

Q: **Practice · 5 marks** Describe two examples of digital rights management used to protect CDs and DVDs, and the case that tested the DVD scheme.

MARKER WORDS: DRM EXAMPLES AND THE CSS CASE

Say it like this:

> **Digital rights management** protects content through methods such as the **Content Scramble System**, an **encryption scheme** that needs **decryption keys** to view a DVD, but **a crack code exists**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
encryption scheme	describing CSS	**SHOULD**
digital rights management	defining or naming DRM	**MUST**
Content Scramble System (CSS)	giving the DVD example	**SHOULD**
decryption keys	explaining how a DVD is viewed	**SHOULD**
digital watermarking	giving the SDMI example	**SHOULD**
Yellow Book vs Red Book	explaining the CD drive vs CD player trick	**SHOULD**
crack code exists	explaining why DRM does not fully work	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

What is CSS (Content Scramble System)?
An encryption scheme for DVD contents. Need decryption keys to view a DVD. Status: in use, but crack code exists.

The three DRM examples for CDs.
Exploit the difference between CD-ROM drives in computers and CD players ("Yellow Book" vs. "Red Book"); encode patterns into audio that translate into annoying sounds when decoded; Secure Digital Music Initiative, which introduces a "digital watermarking" scheme.

The Jon Johansen case in four steps.
Johansen wrote a decryption program for Linux; 2600.com published the code; motion picture studios sued 2600.com and won; Johansen was tried in Norway and found not guilty.

Criticisms of DRM

Slide 19 gives three criticisms and one closing remark.

1. **DRM undermines fair use.** A lock cannot tell a legal copy (time shifting, space shifting, a quote for teaching) from an illegal one. It blocks both, so the user's legal fair use right is lost.
2. **Some DRM schemes track user information.** To control use, some schemes record what you play, read or copy, which is a privacy problem.
3. **DRM protections never expire (unlike copyrights).** A copyright ends after a fixed term and the work then enters the public domain. A technical lock stays on, so the work is never really free to use, even after the legal protection ends.

Closing remark. Many people believe that **any technological "fix" is bound to fail**, because determined people find ways around it. The crack for CSS is the evidence. The slide then asks "The alternative?", which leads to the discussion of tougher laws, open licensing and new business models, and the open questions are in t15.

> **Key point.** DRM, tougher laws and the other answers come from the same two competing goals in t12: reward creators and benefit society. DRM leans too far toward creators. The fair use criticism is the way to say so.

Q: **Practice · 6 marks** **Give three criticisms of digital rights management.**

MARKER WORDS: CRITICISMS OF DRM

Say it like this:

> DRM **undermines fair use**, some schemes **track user information**, and its protections **never expire**, unlike copyrights.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
undermines fair use	giving the first criticism	SHOULD
track user information	raising the privacy criticism	SHOULD
never expire	contrasting DRM with copyright's limited term	SHOULD
technological fix is bound to fail	concluding that DRM alone is not enough	SHOULD
digital rights management	naming the technology you are criticising	MUST

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The three criticisms of DRM.
DRM undermines fair use; some DRM schemes track user information; DRM protections never expire (unlike copyrights).

What do many people believe about technological fixes for IP protection?
Any technological "fix" is bound to fail.

Copyright law, software protection and the DMCA

~35 min to read | High priority | 7 questions

Source: Set 5, slides 20-29

This section follows on from the basics of IP protection and fair use (t12) and the online protection challenges, DRM and landmark cases (t13). It covers how copyright law has grown over time, how punishments have become harsher, what the Digital Millennium Copyright Act (DMCA) did, and the argument over how software should be protected. The open-source answer to that argument is in t15.

US copyright law: a history of expansion

The slides use the United States as the case study. The pattern is simple: each new law covered more kinds of work and lasted longer than the one before.

The first copyright law was passed in **1790**. It gave **28 years** of protection (two terms of 14 years: 14+14) and covered books only. After that, **each subsequent revision broadened the scope of coverage and increased the protection period**.

Law	What it newly covered	Length of protection
1790 Act	Books	28 years (14+14)
1831 Act	Prints and sheet music	42 years
1909 Act	Photos, recordings, movies	56 years
1976 Act	Software and databases	75 years
1998 Act	Music broadcast over the Internet	Author's lifetime + 95 years

Slide 21 is a picture (credited to Tom Bell on Wikipedia) that plots the length of copyright against time. The x-axis is the year a copyright term began (1790 to 2008) and the y-axis is the duration of the term in years (up to 120). Each Act is a coloured band that steps the term up: 28, 42, 56, 75, then about 95 years after the 1998 (Sonny Bono) Act. The legend also lists the 1962-74 Acts, short interim extensions that are not on the text slide. It adds nothing you need to memorise beyond the table.

Two things to notice. First, the **1976 Act** is the one that brought **software and databases** into copyright; this is why the software-protection debate later in this section starts there. Second, the term changed from a fixed number of years to a period tied to the **life of the author**, which is a much longer period.

> **Watch out.** The slide says "author's lifetime + 95 years" for the 1998 Act. Your solution files (and many textbooks) say "life of the author plus 70 years". **beyond the slides** In real US law, life plus 70 years applies to individual authors and 95 years from publication applies to corporate works, which probably explains the two numbers. If a question uses your lecturer's figures, use the slide's. If you want to be safe, write "the life of the author plus several decades (95 years in the slides)".

Criticisms of longer protection

The slides start from the **purpose of copyright stated in the US Constitution** (Article I, Section 8, Clause 8). Congress has the power:

> "to promote the Progress of Science and useful Arts, by securing for **limited times** to Authors and Inventors the **exclusive Right** to their Respective Writings and Discoveries."

The key words are *promote progress* and *limited times*. Copyright exists to help society, and the owner's monopoly is meant to be temporary. This links straight back to the two competing goals of IP protection:

benefit the society (get inventions to the public domain without delay) and reward the creators.

Critics say that making protection longer and longer works against that purpose. The slides give three criticisms.

- **It benefits only the few owners of franchises.** The big winners are companies that own valuable old works. An author who is already dead cannot be encouraged to create anything by an extension.
- **The 1998 Act is called the "Mickey Mouse Protection Act".** The nickname comes from the claim that it was pushed through to stop early Mickey Mouse works from falling out of copyright. beyond the slides The Act is formally the Sonny Bono Copyright Term Extension Act.
- **It will shift progress of science and arts offshore, where copyrights would have expired.** If a work is still locked up at home but free in another country, people in that country can build on it, adapt it and publish it. Creative and scientific progress then moves there.

Notice the shape of the argument: the critics accept that copyright is justified by progress, and then show that longer terms do not serve progress. That is a consequentialist (utilitarian) style of criticism.

Q: Practice · 6 marks **Outline the history of copyright law in the United States and give three criticisms of making the protection period longer.**

MARKER WORDS: COPYRIGHT LAW HISTORY AND LONGER PROTECTION

Say it like this:

> Since the first law in **1790**, each revision of US copyright law has **broadened the scope of coverage and increased the protection period**. Critics say longer protection **benefits only the few owners of franchises** and goes against the Constitution's aim to promote progress by securing rights for **limited times**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
limited times / **limited time**	stating the Constitution's purpose or the limits on ownership	**MUST**
exclusive right	saying what copyright gives the owner	**MUST**
life of the author plus (lifetime + 95 years)	giving the current length of protection	**MUST**
public domain	saying what happens when protection ends	**MUST**
promote the progress of science and useful arts	giving the constitutional purpose	**SHOULD**
broadened coverage and increased protection period	describing the trend of the laws	**SHOULD**
Mickey Mouse Protection Act	criticising the 1998 Act	**SHOULD**
shift progress offshore	giving the third criticism	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

US copyright laws in order, with the protection period of each.
1790: books, 28 (14+14) years. 1831: prints and sheet music, 42 years. 1909: photos, recordings, movies, 56 years. 1976: software and databases, 75 years. 1998: music broadcast over the Internet, author's lifetime + 95 years.

The Constitution's stated purpose of copyright law (Article I, Section 8, Clause 8).
To promote the Progress of Science and useful Arts, by securing for limited times to Authors and Inventors the exclusive Right to their Respective Writings and Discoveries.

Three criticisms of longer copyright protection.
It benefits only the few owners of franchises; the 1998 Act is called the "Mickey Mouse Protection Act"; it will shift the progress of science and arts offshore, where copyrights would have expired.

Harsher punishments

The second way the law has changed is that **punishments for infringement have become much harsher**. This is one of the three "answers to the challenges" of digital copying listed in t13 (prevention through DRM, tracking through watermarking, punishing through extended copyright laws). The slides list three laws.

- **No Electronic Theft (NET) Act, 1997.** Any unauthorised copying with a **retail value of more than $1000** is covered. The important idea is that it applies even when the copier makes *no profit*: before this law, giving copies away free was hard to prosecute. **beyond the slides**
- **Digital Millennium Copyright Act (DMCA), 1998. Many copyright violations become felonies.** A felony is a serious crime (as opposed to a civil wrong or a minor offence).
- **Digital Theft Deterrence and Copyright Damages Improvement Act, 1999.** A **minimum of $750 for each infringement**, and up to **5 years in prison or a $500,000 fine for a first offence**.

Put together, the trend runs: a low threshold for what counts as criminal copying (1997), felony status for many violations (1998), then fixed minimum damages and prison terms (1999). The RIAA lawsuits described in t13 (Jammie Thomas fined $222,000 for sharing 24 songs) are examples of what these laws allow.

The DMCA and its consequences

The Digital Millennium Copyright Act was **passed in 1998 but is still controversial**. Its heart is this rule: **the DMCA makes circumventing copy control a felony, even if it is done for a fair-use purpose.**
"Circumventing copy control" means breaking or bypassing a technical protection such as DRM or the DVD scrambling scheme (CSS, see t13). The usual defence that "this was fair use" does not help once a protection was bypassed. That is why critics say the DMCA and DRM together *undermine fair use* (compare the DRM criticisms in t13 and the fair-use factors in t12).

The slides give four consequences.

1. **Inconsistency between media.** Making a personal copy of a music CD is fair use, but it is illegal to do so for a movie DVD. The reason is that DVDs are protected by CSS, so copying means circumventing it, while music CDs have no such lock.
2. **It is illegal to play a DVD on a GNU/Linux machine.** Early Linux had no licensed DVD player, so users had to use a decryption program, which counts as circumvention. This is the Jon Johansen story from t13.
3. **Online service providers (OSPs) may be held liable, and face severe penalties, for copyright violations by their users.** A website or hosting service can be punished for what its users upload. This

pushes providers to remove content quickly or to monitor users.

4. **The Viacom v. YouTube lawsuit is on-going.** Viacom (a media company) sued YouTube for clips of its programmes uploaded by users. It is the test of how far an OSP is responsible. (The slide was written in 2010 when the case was still running.)

Key point. The DMCA protects the *lock*, not just the work. Even if your use of the work would be legal, breaking the lock is a felony. That is the sentence to remember when asked why the DMCA is controversial.

Q: Practice · 6 marks **Explain why the Digital Millennium Copyright Act is controversial. Give three consequences.**

Q: Practice · 4 marks **List the three US laws since 1997 that made punishments for copyright infringement harsher, and state what each provides.**

MARKER WORDS: HARSHER PUNISHMENTS AND THE DMCA

Say it like this:

> The **DMCA (1998)** makes **circumventing copy control a felony, even for a fair-use purpose**, and **online service providers may be held liable** for copyright violations by their users.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
circumventing copy control	saying what the DMCA forbids	**SHOULD**
felony	describing how serious the penalty is	**SHOULD**
even for fair use	explaining why the DMCA is controversial	**SHOULD**
online service provider liable	giving a consequence for websites	**SHOULD**
No Electronic Theft Act (retail value over $1000)	naming the 1997 law	**SHOULD**
$750 per infringement, 5 years or $500K	giving the 1999 penalties	**SHOULD**
undermines fair use	criticising DMCA and DRM	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The three harsher-punishment laws, with years.
No Electronic Theft Act (1997); Digital Millennium Copyright Act (1998); Digital Theft Deterrence and Copyright Damages Improvement Act (1999).

What does the No Electronic Theft Act (1997) cover?
Any unauthorised copying with retail value over $1000.

What does the DMCA (1998) do?
Many copyright violations become felonies. It makes circumventing copy control a felony, even if it is for fair-use purposes.

Penalties under the Digital Theft Deterrence and Copyright Damages Improvement Act (1999). Minimum $750 for each infringement; up to 5 years in prison or a $500K fine for a first offence.

Four consequences of the DMCA.
A personal copy of a music CD is fair use, but illegal for movie DVDs; it is illegal to play a DVD on a GNU/Linux machine; online service providers may be held liable (with severe penalties) for violations by their users; the Viacom v. YouTube lawsuit is on-going.

Software protection: the two arguments

Slide 11 of t13 asked "should software be copyrighted? should it be patented?". Slide 25 begins the answer by asking *why software should be protected at all*. It gives two kinds of argument, and the difference between them is the difference between the two big ethical theories (see t01).

Utilitarian arguments (consequences)

The slides give a chain of consequences, each one leading to the next:

1. Copying software **reduces software purchases**.
2. That leads to **less income for software makers**.
3. That leads to **lower production of new software**.
4. That leads to **fewer benefits to society**.

The conclusion is that copying should be illegal because the total good for society falls. The argument judges the law by its outcomes, so it is **utilitarian** (consequentialist).

Rights-based arguments (deontological)

- **Programming is hard work that only a few can do.**
- **Programmers should be rewarded for their labour.**
- **They ought to be able to own their programs.**

This argument ignores outcomes. It says the programmer has a *right* to what their labour produced, whatever the effect on society. beyond the slides This is the same thought as the labour theory of property (a person owns what they mix their work into), and it is a duty or rights style of reasoning. Use the slide wording first and add the theory name as a bonus.

The basic dilemma

The slides do not use the word "dilemma" here, but your question bank asks for it. beyond the slides The dilemma is this: giving the programmer ownership and control rewards labour and encourages new software, but it also lets the owner restrict who may use, copy or improve the software. Software can be copied at almost no cost and without taking anything from the original owner. Ownership therefore keeps useful software away from people who could benefit, and it can give owners monopoly power and high prices. So society must balance **the owner's rights and incentive** against **the public's access, affordability and progress**. Your solution files phrase it as *monopoly power* and *access and affordability*.

Q: **QB D2 · 6 marks** What is the consequentialist justification for laws that give ownership and control of software to individuals or corporations (proprietary software)? What is the basic dilemma in giving individuals the ownership and control of software they wrote?

Q: **2016 Q4A · 8 marks; also CAT 1 2021-22 Q3b · utilitarian arguments on software property rights** There are a number of arguments for and against strong property rights over computer software. Provide two utilitarian-style arguments for each side. [2016 wording reconstructed from garbled scan: "construct two utilitarian-style arguments supporting, and two opposing, strong property rights in software"]

MARKER WORDS: WHY SOFTWARE SHOULD BE PROTECTED

Say it like this:

> The **utilitarian argument** is that **copying software reduces software purchases**, leading to **less income for software makers**, **lower production of new software** and **fewer benefits to society**. The **rights-based argument** is that programming is **hard work** and programmers **should be rewarded for their labour** and be able to **own their programs**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
utilitarian argument / **consequences**	labelling the first argument	**MUST**
incentive	explaining why ownership encourages new software	**MUST**
monopoly	giving the cost of strong ownership	**MUST**
dilemma (rights vs public access)	answering the "basic dilemma" part	**MUST**
less income, lower production, fewer benefits	laying out the chain of consequences	**SHOULD**
rights-based (hard work, rewarded for labour)	giving the second kind of argument	**SHOULD**
own their programs	stating the programmer's claim	**SHOULD**
fewer benefits to society	closing the utilitarian chain	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The utilitarian argument for software protection (the chain).
Copying software reduces software purchases; leading to less income for software makers; leading to lower production of new software; leading to fewer benefits to society.

The rights-based arguments for software protection.
Programming is hard work that only a few can do; programmers should be rewarded for their labor; they ought to be able to own their programs.

Software copyrights

Software was brought into copyright by the **1976 revision**. The slides explain what exactly is protected, using the copyright rule that applies to every work.

- **Expression of an idea, not the idea itself.** You may not copy the way the program is written, but nobody owns the idea (for example "a program that sorts names"). Another programmer may write a different program that does the same job.
- **Object code, not source code.** The copyright covers the machine-readable program that is sold. **Companies treat source code as a trade secret** (see trade secrets in t12) and keep it hidden, so the two protections work together.

Three examples of violations from the slides:

1. Copying a program to give or sell to someone else.
2. Preloading a program onto the hard disk of a computer being sold (a shop installs software on computers without a licence).
3. Distributing a program over the Internet.

These all breach the owner's copyright rights of *reproduction* and *distribution* (see the five rights in t12). The question on the two forms of legal protection for software, 2016 Q4B (**6 marks** :

> EXAM WORDING
> What two forms of legal property protection are most frequently applied to software? List two or three of their key characteristics.

), is answered in t12; here we add the debate over patents.

Software patents

The slides ask "software patents?" with a question mark because this is the most disputed form of software protection.

When is software patentable? The slide's rule: **software is patentable if it contains a mathematical formula and implements it in a structure which performs a patentable function.** Patents are **not to be given for things that are obvious or already in common use**. The trouble is that **it is hard to determine what is "obvious"**, and the result is that **some bad patents have been issued**.

Examples of software patents (slide 28):

Category	Examples
Data compression schemes	GIF, JPEG, MP3, RSA
Internet tools	Pay-per-click ads, the browser plug-in
User interfaces	The progress bar, force feedback (in game controllers)

Remember a patent lasts 20 years and requires a public, detailed description (t12). Ideas such as the progress bar were seen by many as obvious, which is the main complaint below.

Criticisms of software patents

Slide 29 gives four.

1. **The cost is too high.** Getting and defending a patent is expensive, so small developers and individuals lose out to large firms with lawyers.
2. **Traditional copyright has provided sufficient protection.** Software was already protected by copyright (and trade secrets) since 1976; a patent adds a second, stronger layer that is not needed.
3. **Most software patents cover trivial inventions or trivial extensions of existing technologies.** This is the "obvious" problem again. Trivial patents let owners block others for 20 years without real invention.
4. **Lack of patent application disclosure: patent applications are published 18 months after they are filed.** A patent exists to publish the invention, but in software the world moves so fast that 18 months of secrecy hides ideas while others unknowingly build the same thing and may then be accused of infringing.

Q: **2016 Q4C · 6 marks; also exam 2021-22 solution file Q5d** **In a few sentences, explain why software poses a challenge to these traditional forms of intellectual ownership.**

Q: **Practice · 8 marks** **State what can and cannot be copyrighted in software, give three examples of copyright violation, and outline four criticisms of software patents.**

MARKER WORDS: SOFTWARE COPYRIGHT AND PATENTS

Say it like this:

> Software copyright protects the **expression of an idea, not the idea itself**, and the **object code, not the source code**. Software is patentable if it implements a **mathematical formula in a structure that performs a patentable function**, but **it is hard to determine what is obvious**, so **some bad patents have been issued**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
traditional copyright has provided sufficient protection	criticising software patents	**SHOULD**
obvious (not obvious, not in common use)	saying when a patent must be refused	**MUST**
exclusive right	saying what a patent or copyright gives	**MUST**
expression of an idea, not the idea itself	saying what copyright covers	**SHOULD**
object code, not source code / trade secret	explaining what is copyrighted	**SHOULD**
trivial inventions	criticising patents	**SHOULD**
published 18 months after filing	giving the disclosure criticism	**SHOULD**
bad patents	describing the result of the "obvious" problem	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

What gets copyrighted in software?
The expression of an idea, not the idea itself; the object code, not the source code (companies treat source code as a trade secret).

Three examples of software copyright violation.
Copying a program to give or sell to someone else; preloading a program onto the hard disk of a computer being sold; distributing a program over the Internet.

When is software patentable?
If it contains a mathematical formula and implements it in a structure which performs a patentable function. Patents are not given for things that are obvious or already in common use.

Software patent examples.
Data compression schemes: GIF, JPEG, MP3, RSA. Internet tools: pay-per-click ad, plug-in browser. User interfaces: progress-bar, force-feedback (in game controller).

Four criticisms of software patents.
Cost is too high; traditional copyright has provided sufficient protection; most software patents cover trivial inventions or trivial extensions of existing technologies; lack of patent application disclosure (applications are published 18 months after they are filed).

Open source and Creative Commons

~35 min to read | Medium priority | 7 questions

Source: Set 5, slides 30-37

This section finishes the intellectual property topic. The earlier sections covered what IP is and how it is protected (IP, protection types, copyright and fair use), why protection is hard online and DRM (Protection challenges, DRM and the landmark cases), and copyright law, software copyright and patents (Copyright law, software protection and the DMCA). Those sections describe the *proprietary* model: the owner keeps control and the law enforces it. This section covers the opposite answer: giving rights away on purpose, using licences.

Open-source software and copyleft

The slides introduce open-source software as the **opposite answer to proprietary software**. Proprietary software is sold or licensed by an owner who keeps the source code secret and restricts copying, modifying and redistributing. Open-source software removes those restrictions. The slide lists four features:

- **No restrictions preventing others from selling or giving away the software.** Anyone who has a copy may sell it or give it away.
- **Source code is included in the distribution.** The human-readable instructions come with the program, not only the compiled version. This is what makes study and change possible.
- **No restrictions preventing others from modifying the source code.** Users may fix bugs, add features or adapt the program.
- **No restrictions regarding how people can use the software.** The licence does not limit the purpose, user or field of use.

The slide then adds **copyleft**, with the General Public Licence (GPL) as the example. Copyleft means: *free to modify and redistribute, as long as the same rights apply to everyone receiving the software.* The author does not give up copyright. The author uses copyright law to force every later user to pass on the same freedoms.

If you take GPL code, improve it and distribute it, you must distribute your version under the same terms, with source code. Nobody can take the shared work and make it proprietary.

The word is a pun on copyright: copyright restricts, copyleft keeps the work free. Open source is therefore not the same as having no licence or no copyright. The code is still owned. The owner chooses a licence that grants broad permissions.

beyond the slides The Free Software Foundation (Richard Stallman, GNU project, 1980s) states the idea as four freedoms: to *run* the program for any purpose, to *study* and change it (needs source code), to *redistribute* copies, and to *distribute modified versions*. The four bullets on the slide match these. Famous examples are Linux, Apache, Firefox and LibreOffice. The Open Source Initiative later promoted the term "open source" with a business-friendly emphasis on practical benefits rather than on freedom as a moral aim. Not all open-source licences are copyleft: permissive licences such as MIT and BSD allow anyone to reuse the code, even inside proprietary products, without passing on the same terms. The GPL is the strong copyleft example the lecturer chose. "Free" here means freedom, not price: open-source software may be sold.

> **Key point.** Open source works through copyright, not against it. The licence is a legal tool. Copyleft uses the owner's right to impose conditions, and the condition is "share alike".

MARKER WORDS: OPEN SOURCE AND COPYLEFT

Say it like this:

> Open-source software is the **opposite answer to proprietary software**: the **source code is included** and there are **no restrictions** on selling, giving away, modifying or using it. **Copyleft** (for example the **GPL**) lets people modify and redistribute it **as long as the same rights apply to everyone receiving the software**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
proprietary software	contrasting open source with the owner-controlled model	**MUST**
open-source licences	explaining how open source challenges traditional ownership and licensing	**SHOULD**
source code included in distribution	saying what makes modification possible	**SHOULD**
no restrictions on modifying	listing the features of open source	**SHOULD**
copyleft	describing share-alike conditions	**SHOULD**
GPL	giving an example of copyleft	**SHOULD**
same rights apply to everyone receiving the software	defining copyleft	**SHOULD**

▶ **Recall drill: write the definition from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The four features of open-source software (slide 30).
No restrictions preventing others from selling or giving away the software; source code included in distribution; no restrictions preventing others from modifying the source code; no restrictions regarding how people can use the software.

Define copyleft (e.g. GPL).
Free to modify and redistribute as long as the same rights apply to everyone receiving the software.

How does open-source software relate to proprietary software?
It is the opposite answer to proprietary software.

Creative Commons licences

The slides say that in **2001** a group of professionals created the **non-profit corporation Creative Commons** to provide **copyright licences free of charge**. The idea is that an author, not the law's default, **decides the extent of IP protection**. Under normal copyright everything is "all rights reserved" automatically. A Creative Commons (CC) licence lets the author say "some rights reserved" without hiring a lawyer. The slide gives two examples:

- **Free to copy, modify and distribute, but no commercial use.**
- **Free to copy and distribute, but no modification or commercial use.**

It also notes that **many Wikipedia pictures are protected by CC licences**. Slides 32 and 33 are titled "Creative Commons (cont.)" and are two screenshots. Slide 32 lists the six Creative Commons 3.0 licences with their icons: Attribution; Attribution-NoDerivs; Attribution-NonCommercial-NoDerivs; Attribution-NonCommercial; Attribution-NonCommercial-ShareAlike; Attribution-ShareAlike. Slide 33 is the human-readable deed for Attribution-NonCommercial-ShareAlike 3.0 Unported. It says: "You are free to **Share** (copy, distribute and transmit the work) and to **Remix** (adapt the work), under the conditions **Attribution**, **Noncommercial** and **Share Alike**." The meaning of each condition is in the table below, which is from general knowledge.

beyond the slides **The four licence elements.** Every CC licence is built from these:

Element	Symbol	What it means
Attribution (BY)	Person icon	Others may use the work only if they credit the author. Present in all six standard licences.
NonCommercial (NC)	Crossed dollar sign	The work may not be used for commercial purposes.
NoDerivatives (ND)	Equals sign	The work may be shared but not changed, remixed or adapted.
ShareAlike (SA)	Circular arrow	Adaptations must be released under the same licence. This is the copyleft idea applied to creative works.

beyond the slides **The six standard licences**, from most to least open:

Licence	Credit required	Commercial use	Modification	Same licence on adaptations
CC BY	Yes	Allowed	Allowed	No
CC BY-SA	Yes	Allowed	Allowed	Yes
CC BY-NC	Yes	Not allowed	Allowed	No
CC BY-NC-SA	Yes	Not allowed	Allowed	Yes
CC BY-ND	Yes	Allowed	Not allowed	Not applicable
CC BY-NC-ND	Yes	Not allowed	Not allowed	Not applicable

The slide's first example (copy, modify, distribute, no commercial use) is CC BY-NC. The second (copy and distribute only, no modification, no commercial use) is CC BY-NC-ND. There is also CC0, a "public domain dedication" in which the author waives rights as far as the law allows. Each licence has three layers: a plain-language summary, the legal text, and machine-readable code so search engines can find CC-licensed work. CC licences apply to creative works such as text, photographs, music and video. They are not designed for software, where the GPL and similar licences are used.

> **Watch out.** A CC licence does not abandon copyright. The author still owns the work and the licence is legally binding. If you break its terms, for example by using a NC image in an advert, you lose the permission and can be sued for infringement. The licence is also irrevocable for those who already received the work under it.

MARKER WORDS: CREATIVE COMMONS

Say it like this:

> **Creative Commons** is a **non-profit corporation** founded in **2001** to provide **copyright licences free of charge**. It **allows the author to decide the extent of IP protection**, for example free copying and distribution but **no modification or commercial use**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
non-profit corporation	saying what Creative Commons is	**SHOULD**
copyright licences free of charge	saying what it provides	**SHOULD**
author decides the extent of IP protection	explaining why CC exists	**SHOULD**
no commercial use	giving an example of a restriction an author can choose	**SHOULD**
no modification	giving the second example	**SHOULD**
Wikipedia pictures	giving a real-world example of CC use	**SHOULD**

▶ **Recall drill: write the definition from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

What is Creative Commons (slide 31)?
A non-profit corporation created in 2001 by a group of professionals to provide copyright licences free of charge. It allows the author to decide the extent of IP protection.

The two CC examples on the slide.
Free copy, modify and distribute, but no commercial use; free copy and distribute, but no modification or commercial use.

The four CC licence elements (beyond the slides).
Attribution (BY); NonCommercial (NC); NoDerivatives (ND); ShareAlike (SA).

Benefits and criticisms of open source

Benefits (slide 34)

- **Programs belong to the entire community.** No single firm owns the program or can withdraw it.
- **Eliminates the tension between obeying the law and helping others.** Under proprietary licences, sharing a program with a friend is illegal copying. Under open source, sharing is legal and encouraged. This answers the dilemma of proprietary software, where the law forbids what feels like a neighbourly act.
- **Gives everyone the opportunity to improve the program.** Anyone can read the code, find bugs and contribute fixes.
- **New versions appear more frequently.** Many contributors work in parallel, so releases come faster.
- **Shifts focus from manufacturing to service.** Copies are free, so firms earn from support, installation, training and customisation, not from selling copies. Red Hat is the standard beyond the slides example.

Criticisms (slide 35)

- **Without an "owner", incompatible versions may arise.** Anyone may modify and redistribute, so the program can split into versions that do not work together (forking). No authority sets the standard.
- **Without a critical mass of developers, quality can be poor.** The "many eyes" benefit only works for popular projects. Small projects can be abandoned or buggy.
- **Relatively weak graphical user interface.** Volunteers like to build engines and tools. Polished, easy interfaces need designers and usability testing, which are costly.
- **Poor mechanism for stimulating innovation: no companies will spend billions on new programs.** This is the consequentialist argument for proprietary software (covered in t14): without the chance to earn money from sales, big investments will not be made. Open source tends to copy and improve existing ideas.

beyond the slides Evidence is mixed. Linux, Apache and Android run much of the world's servers and phones, and big companies pay staff to work on open source. That weakens the "no innovation" claim. Security is also debated: public code can be audited, but attackers can read it too. For a student in Uganda, open source also means free tools and no licence bills, which is a strong practical benefit.

MARKER WORDS: BENEFITS AND CRITICISMS OF OPEN SOURCE

Say it like this:

> Open source benefits include that **programs belong to the entire community**, it **eliminates tension between obeying the law and helping others**, everyone can **improve the program**, **new versions appear more frequently** and the focus **shifts from manufacturing to service**. Its weaknesses are **incompatible versions**, **poor quality without a critical mass of developers**, a weak interface and a **poor mechanism for stimulating innovation**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
stimulating innovation	arguing that proprietary rights reward investment, or criticising open source	**SHOULD**
community	saying who owns open-source programs	**MUST**
service	explaining how open-source firms make money	**MUST**
tension between obeying the law and helping others	explaining a benefit of open source	**SHOULD**
critical mass of developers	explaining why quality can be poor	**SHOULD**
incompatible versions	explaining the cost of having no owner	**SHOULD**
weak graphical user interface	giving a usability criticism	**SHOULD**

▶ **Recall drill: write the benefits sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The six Creative Commons 3.0 licences (slide 32).
Attribution; Attribution-NoDerivs; Attribution-NonCommercial-NoDerivs; Attribution-NonCommercial; Attribution-NonCommercial-ShareAlike; Attribution-ShareAlike.

The five benefits of open source (slide 34).
Programs belong to the entire community; eliminates tension between obeying law and helping others; gives everyone opportunity to improve program; new versions of programs appear more frequently; shifts focus from manufacturing to service.

The four criticisms of open source (slide 35).
Without an owner, incompatible versions may arise; without critical mass of developers, quality can be poor; relatively weak graphical user interface; poor mechanism for stimulating innovation, as no companies will spend billions on new programs.

Comparison: proprietary, open source and Creative Commons

Feature	Proprietary	Open source (e.g. GPL)	Creative Commons
Typical subject	Commercial software, films, music	Software	Creative works: text, images, music, video
Who decides the terms	The owner or publisher	The author, by choosing a licence	The author, by choosing from the CC menu
Source code	Kept secret	Included in distribution	Not usually relevant
Copying and redistribution	Restricted or forbidden	Allowed	Allowed within the licence terms
Modification	Forbidden	Allowed	Depends: allowed unless ND
Commercial use	Owner only, or by paid licence	Allowed	Depends: forbidden if NC
Share-alike condition	Not applicable	Yes for copyleft (GPL)	Yes if SA is chosen
Main legal basis	Copyright, patents, trade secrets, contracts	Copyright used through a licence	Copyright used through a licence
Revenue model	Selling copies or licences	Service, support, customisation	Attribution, goodwill; some NC works earn from paid commercial licences
Main strength	Rewards investment and innovation	Community improvement, free sharing	Simple, free, legal sharing with clear rules
Main weakness	Cost, tension between law and helping others	Incompatible versions, weak interfaces, innovation funding	Licence misuse, NC ambiguity, irrevocable once granted

The comparison column on Creative Commons and the "revenue model" and "weakness" rows are partly **beyond the slides**. The slides give the open-source and CC features and the criticisms listed above.

Practice: open source, copyleft and Creative Commons

Q: Practice · 6 marks What is open-source software? Explain the term copyleft.

Q: Practice · 9 marks Outline the benefits and the criticisms of open-source software.

Q: Practice · 4 marks What is Creative Commons? Give two examples of the licences it offers.

Discussion questions with model answers

These four questions are on slides 36 and 37. They are discussion questions, so a good answer argues both sides and ends with a verdict. They are likely exam material because the lecturer asks "give arguments for each side".

Q: Set 5 slide 36 · discussion A software service company A has a client C. C uses a software S made by company B. Company A made a copy of S, not to resell, but to provide service to C. Company B sued, and company A argues that the copying was a fair use. Give arguments for each side. Which side do you think should win?

Q: Set 5 slide 36 · discussion Consider applying the fair-use guidelines to making a video of oneself lip-synching to a popular song and posting the video on a social-networking site. Do you think it is ethical to do this?

Q: Set 5 slide 37 · discussion Which is more likely to be effective in protecting intellectual property in digital media (e.g. CDs, DVDs): tougher copyright laws, or new technologies incorporating more sophisticated anti-copying measures?

Q: Set 5 slide 37 · discussion As the copying of digital texts, audio, and video grows cheaper, the restrictions needed to enforce copyright grow more expensive and invasive. Does copyright have a future in the digital age? Or are new technologies steadily making it obsolete?

MARKER WORDS: THE DISCUSSION QUESTIONS

Say it like this:

> To judge **fair use**, courts consider the **purpose and character of use**, the **nature of the work**, the **amount of the work copied** and the **effect on the commercial market**. Protection of digital media needs **prevention** (DRM), **tracking** (watermarking) and **punishing** (law), but each has costs.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
licence	explaining what the client or the open-source author has or grants	**SHOULD**
purpose and character of use	applying fair use to a case	**SHOULD**
effect on the commercial market	deciding whether a copy hurts the owner	**SHOULD**
amount of work copied	noting that whole-work copying counts against fair use	**SHOULD**
anti-copying measures	discussing DRM against law	**SHOULD**
invasive	criticising heavy enforcement	**SHOULD**

▶ **Recall drill: write the fair-use factors from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The four fair-use factors courts consider (slide 8, used in the discussion questions).
Purpose and character of use; nature of the work; amount of work being copied; effect on the commercial market for the work.

The three ways to answer the digital IP challenges (slide 12).
Prevention (digital IP protection techniques, e.g. DRMs); tracking (digital watermarking); punishing (extending copyright laws).